

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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Contents

1	Overview	1
2	Load Data	2
3	Preprocessing	2
4	Primary: Within-Condition Slot Slope-Interaction Asymmetry	2
5	Bootstrap CIs and Effect Sizes on A_influence_story	4
6	Planned Contrast: HA vs (HH + AA)/2	4
7	Directional Stability of Signed Novelty-Resonance Slope Asymmetry	4
8	Length-Controlled Variant of A_influence	5
9	Supplemental: General Condition-Level Comparison	5
10	Supplemental: Pooled Novelty-to-Resonance Relationship	8
11	Supplemental: Role-Level View	10
12	Summary	10

1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions

- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|,$$

extracted via **emtrends**. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff’s δ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in analysis/human-ai/novelty.Rmd and is not duplicated here. Starter-mechanism analysis is deferred.

Article-use note. The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	80	640
human-ai	100	692
human-human	36	288

3 Preprocessing

Slot-level rows: 3080

Metric rows: 9240

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from $\text{resonance} \sim \text{surprise} *$ slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	0.913	0.894	-0.019	0.041	-0.458	0.647	0.019	
Human-Human	0.971	0.985	0.014	0.056	0.242	0.809	0.014	
AI-AI	0.906	0.941	0.035	0.033	1.047	0.295	0.035	

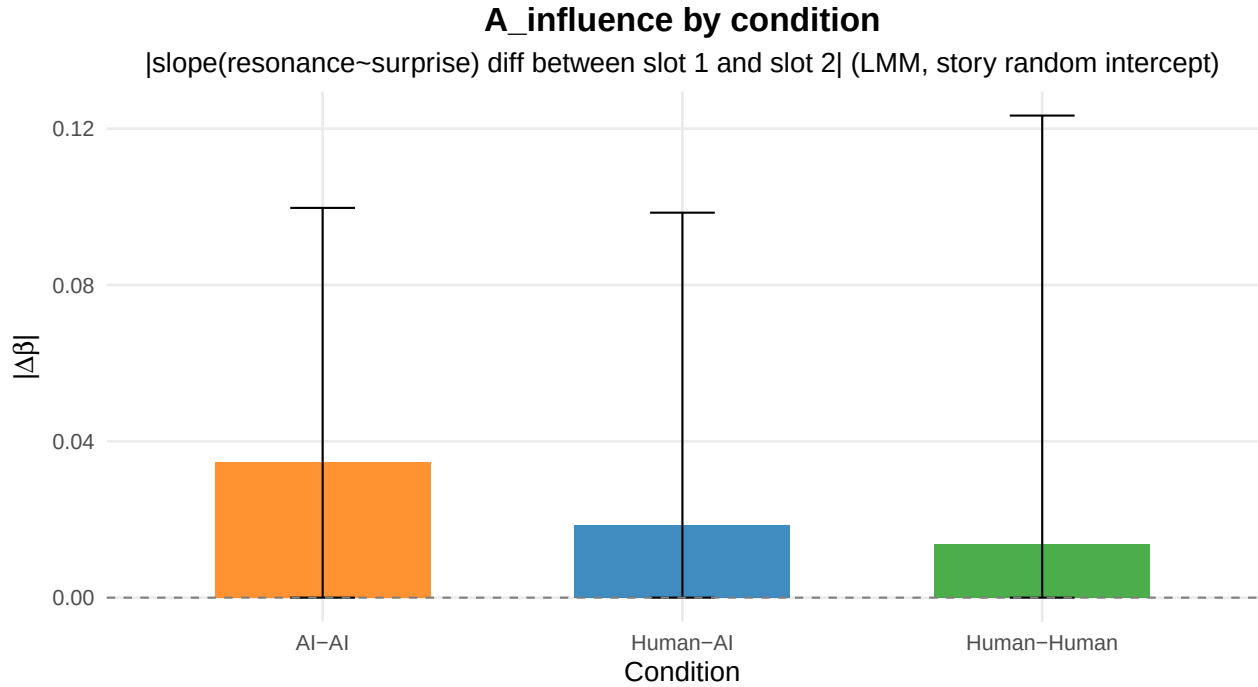


Table 3: Pooled model: |slope_slot1 - slope_slot2| (unsigned) by condition

condition	A_influence	se
Human-AI	0.025	0.038
Human-Human	0.016	0.057
AI-AI	0.039	0.037

Table 4: Robustness: per-story |slope_1 - slope_2| (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	100	0.418	0.519
Human-Human	36	0.382	0.456
AI-AI	80	0.343	0.429

```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 1.8218, df = 2, p-value = 0.4022
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
```

```
## data: per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
##           Human-AI Human-Human
## Human-Human 0.67      -
## AI-AI       0.57      0.57
##
## P value adjustment method: BH
```

5 Bootstrap CIs and Effect Sizes on A_influence_story

Table 5: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	100	0.519	0.439	0.602
Human-Human	36	0.456	0.363	0.554
AI-AI	80	0.429	0.359	0.504

Table 6: Pairwise Cliff’s delta on per-story A_influence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	100	36	0.049	negligible
Human-AI vs AI-AI	100	80	0.109	negligible
Human-Human vs AI-AI	36	80	0.102	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: per-story A_influence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.076	0.052	1.475	171.137	0.142	-0.026	0.178

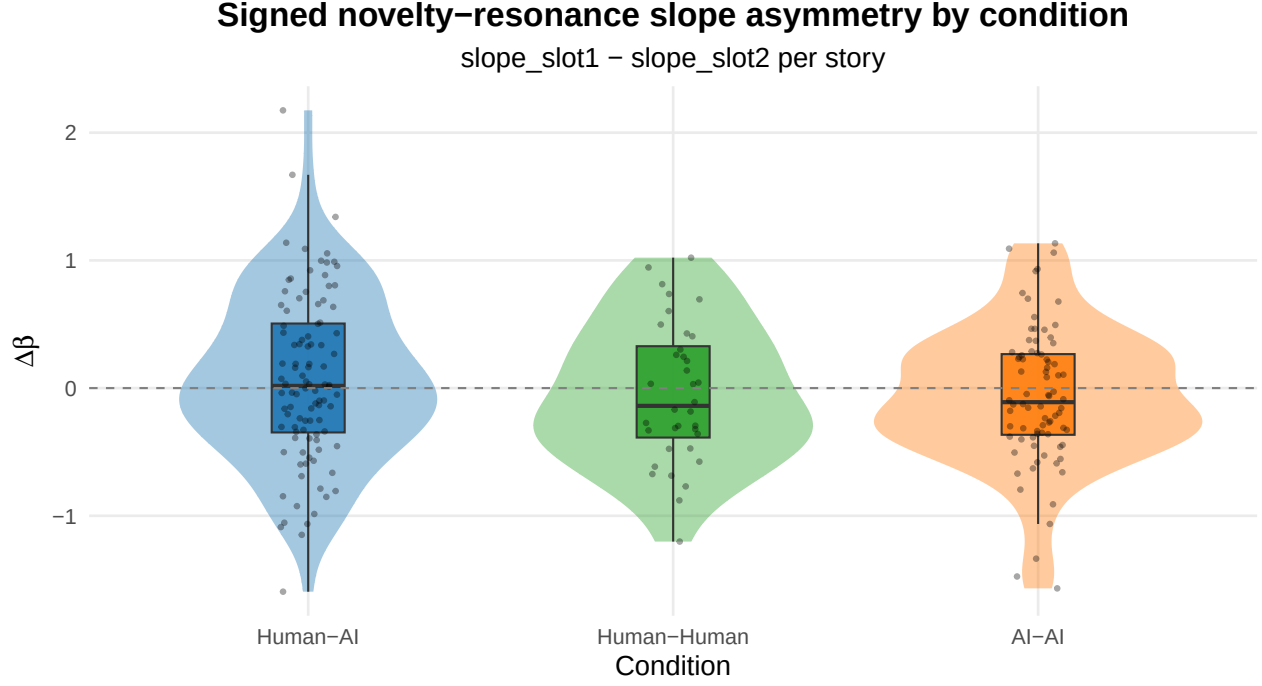
7 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned A_influence is similar.

Table 8: Per-condition directional stability of signed novelty-resonance asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	100	0.071	0.661	0.107	0.510	0.285
Human-Human	36	-0.044	0.547	0.081	0.528	0.632
AI-AI	80	-0.065	0.543	0.120	0.575	0.287

Brown-Forsythe variance-equality p = 0.1996



8 Length-Controlled Variant of $A_{\text{influence}}$

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

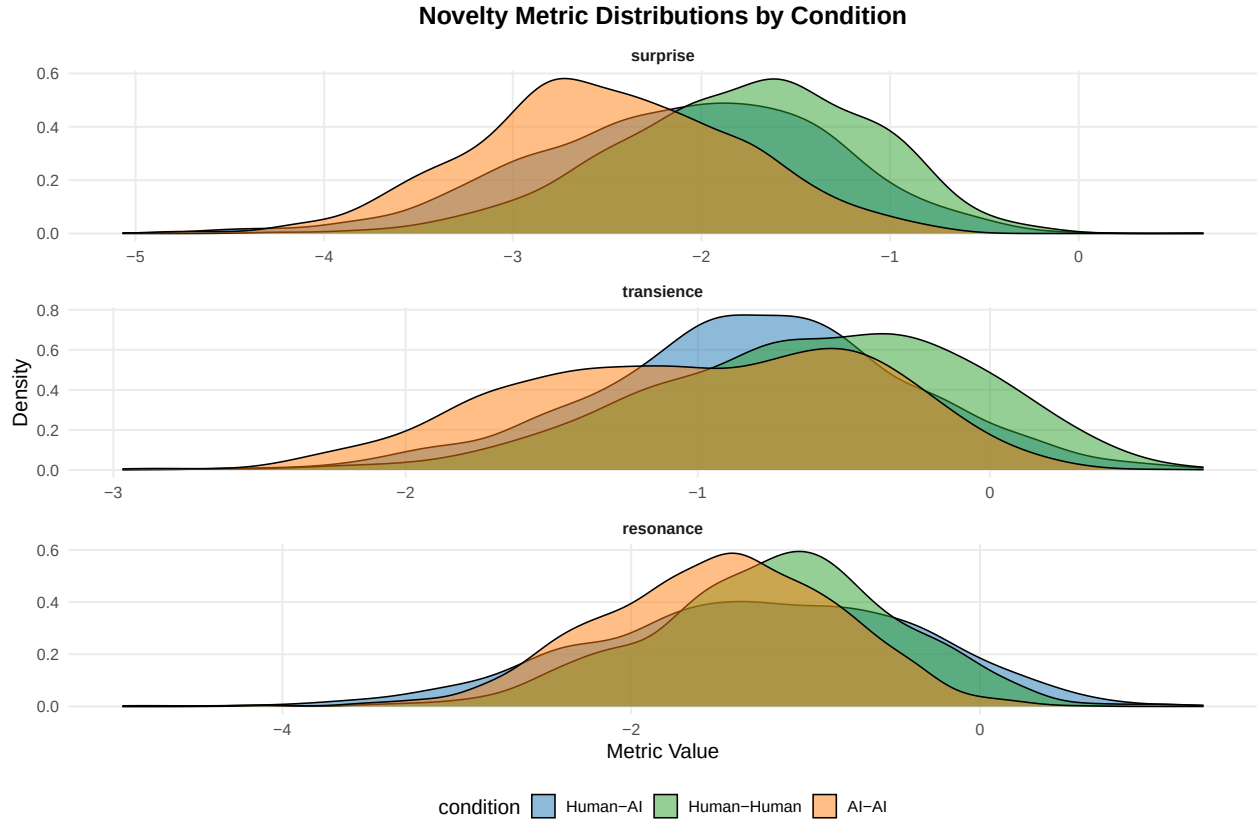
Table 9: Length-controlled $|\text{slope_slot1} - \text{slope_slot2}|$ (resonance ~ surprise * slot + nwords)

condition	$A_{\text{influence_lc}}$	se	p
Human-Human	0.013	0.056	0.821

9 Supplemental: General Condition-Level Comparison

Table 10: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	100	1284	-2.112	0.781
Human-AI	transience	100	1284	-0.815	0.535
Human-AI	resonance	100	1284	-1.298	0.922
Human-Human	surprise	36	557	-1.784	0.671
Human-Human	transience	36	557	-0.609	0.550
Human-Human	resonance	36	557	-1.176	0.719
AI-AI	surprise	80	1239	-2.507	0.690
AI-AI	transience	80	1239	-1.004	0.594
AI-AI	resonance	80	1239	-1.503	0.679



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6395
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9832 -0.6152  0.0327  0.6537  3.9062
##
## Random effects:
```

```

## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1098  0.3313
## Residual          0.4173  0.6460
## Number of obs: 3080, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -2.11384    0.03773 224.48797 -56.029 < 2e-16 ***
## conditionHuman-Human  0.32246    0.07226 211.79966  4.462 1.32e-05 ***
## conditionAI-AI      -0.40189    0.05597 214.82447 -7.181 1.12e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.522
## condtnAI-AI -0.674  0.352
##
## contrast          estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human)  -0.322 0.0723 Inf  -4.462 <.0001
## (Human-AI) - (AI-AI)         0.402 0.0560 Inf   7.181 <.0001
## (Human-Human) - (AI-AI)      0.724 0.0742 Inf   9.760 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4110.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.1546 -0.6174  0.0327  0.6618  3.3502
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1251  0.3537
## Residual          0.1880  0.4336
## Number of obs: 3080, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -0.81371    0.03739 219.07320 -21.763 < 2e-16 ***
## conditionHuman-Human  0.19546    0.07219 213.32744  2.708 0.007326 **
## conditionAI-AI      -0.20314    0.05580 214.70484 -3.640 0.000341 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H

```

```

## cndtnHmn-Hm -0.518
## condtnAI-AI -0.670 0.347

## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -0.195 0.0722 Inf -2.708 0.0203
## (Human-AI) - (AI-AI) 0.203 0.0558 Inf 3.640 0.0008
## (Human-Human) - (AI-AI) 0.399 0.0744 Inf 5.361 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 7331.7
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.2079 -0.6319 0.0380 0.6805 3.2778
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.01833 0.1354
## Residual 0.61489 0.7841
## Number of obs: 3080, groups: conversation_id, 216
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -1.29848 0.02574 239.30515 -50.448 < 2e-16 ***
## conditionHuman-Human 0.12343 0.04771 198.13639 2.587 0.0104 *
## conditionAI-AI -0.20333 0.03726 207.72485 -5.457 1.37e-07 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H
## cndtnHmn-Hm -0.540
## condtnAI-AI -0.691 0.373

## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -0.123 0.0477 Inf -2.587 0.0290
## (Human-AI) - (AI-AI) 0.203 0.0373 Inf 5.457 <.0001
## (Human-Human) - (AI-AI) 0.327 0.0484 Inf 6.756 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests

```

10 Supplemental: Pooled Novelty-to-Resonance Relationship

```

## Novelty-to-resonance model:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [

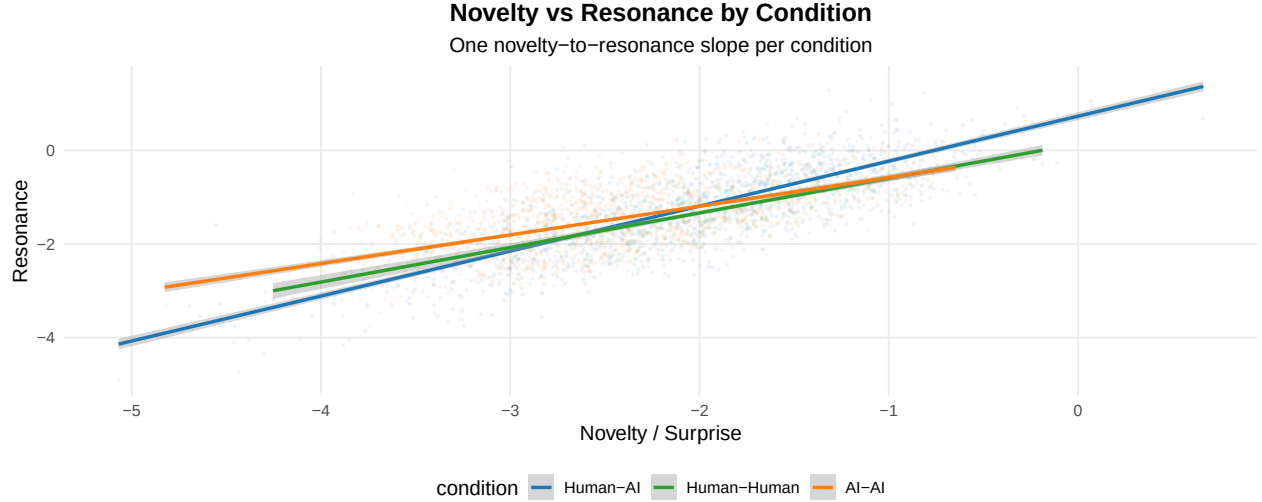
```



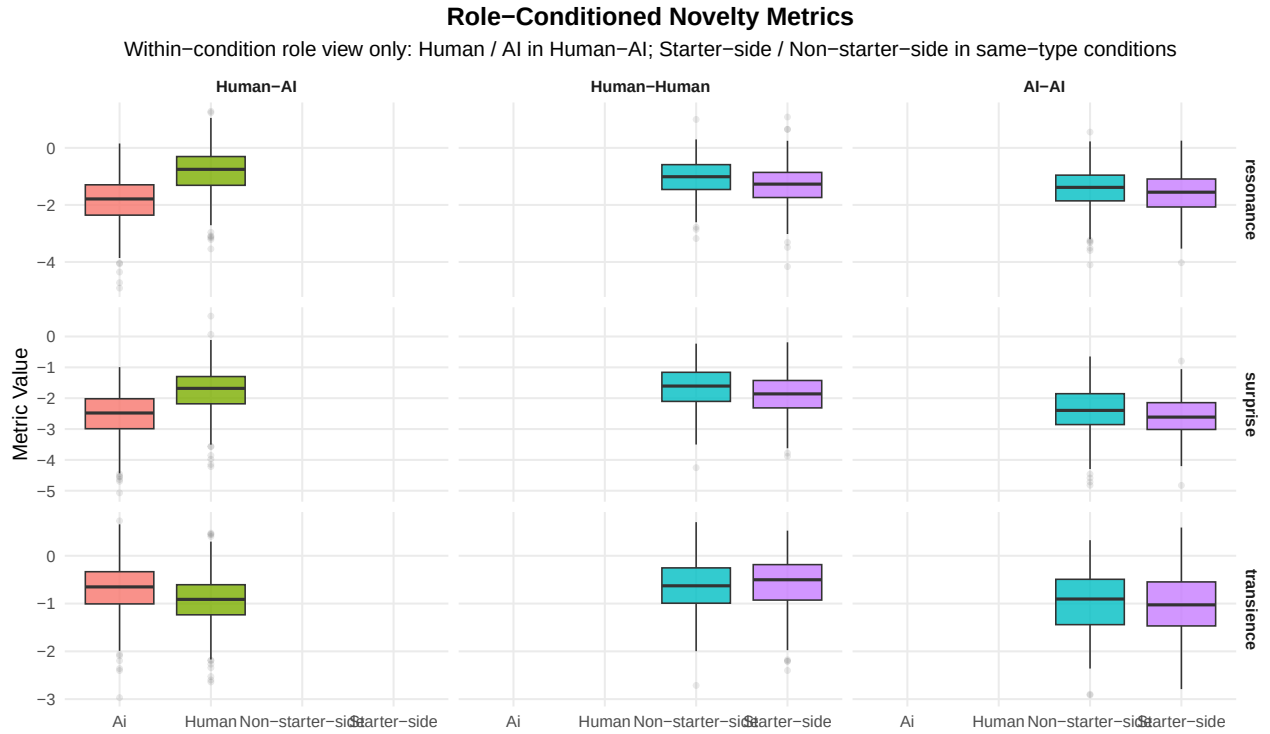
```

## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
## Data: df_slots
##
## REML criterion at convergence: 4103.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.3670 -0.6557 -0.0237  0.6100  4.3144
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1133   0.3366
## Residual                0.1879   0.4335
## Number of obs: 3080, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.87853    0.05014  725.66214  17.521 < 2e-16
## surprise          1.03065    0.01662 2955.35827  62.013 < 2e-16
## conditionHuman-Human -0.30296    0.09672  687.01223  -3.132 0.00181
## conditionAI-AI      -0.11384    0.08498 1001.83698  -1.340 0.18069
## surprise:conditionHuman-Human -0.05442    0.03635 3024.50046  -1.497 0.13441
## surprise:conditionAI-AI -0.13080    0.02778 3046.10435  -4.708 2.62e-06
##
## (Intercept)          ***
## surprise              ***
## conditionHuman-Human **
## conditionAI-AI
## surprise:conditionHuman-Human
## surprise:conditionAI-AI ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise      0.701
## cndtnHmn-Hm -0.518 -0.363
## condtnAI-AI -0.590 -0.413 0.306
## srprs:cnH-H -0.320 -0.457 0.699 0.189
## srprs:AI-AI -0.419 -0.598 0.217 0.776 0.274
##
## Condition-specific novelty-to-resonance slopes:
## condition surprise.trend SE df asymp.LCL asymp.UCL
## Human-AI      1.031 0.0166 Inf 0.998 1.063
## Human-Human    0.976 0.0323 Inf 0.913 1.040
## AI-AI          0.900 0.0223 Inf 0.856 0.943
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95

```



11 Supplemental: Role-Level View



12 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from $\text{resonance} \sim \text{surprise} * \text{slot} + (1|\text{conversation_id})$. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story $A_{\text{influence}}$
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

- length-controlled re-fit (+ `nwords`) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled `condition` LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in `analysis/human-ai/novelty.Rmd`)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-05 11:23:53.141425